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General Comment

Please find attached comments from STM on the AI Action Plan.

Attachments

US Artificial Intelligence (AI) Action Plan March 2025

STM Advancing trusted research

March 15, 2025

Faisal D'Souza
National Coordination Office (NCO)
Networking and Information Technology Research and Development (NITRD)
National Science Foundation
2415 Eisenhower Avenue
Alexandria, VA 22314

RE: STM Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan (90 FR 9088)

Dear Mr. D'Souza:

STM, the International Association of Scientific, Technical and Medical Publishers welcomes the opportunity to provide input on the development of an AI Action Plan, as published in the Federal Register on February 6, 2025.

STM represents non-profit scientific societies, university presses, and commercial publishers. Our members play a vital role in disseminating high-quality, peer-reviewed research that drives scientific discovery and innovation. We are deeply invested in ensuring that AI technologies are developed and deployed in a manner that promotes discovery and innovation while ensuring the integrity and reliability of the information provided by the technology, protecting intellectual property, and fostering public trust.

We appreciate the Trump Administrations commitment to build on America's leadership in AI development to promote human flourishing, economic competitiveness, and national security.

STM and its member publishers understand that AI has the potential to radically change the way we work, learn, do research, and deal with information, and STM members are already deploying AI to support accurate, transparent, and reliable research and discovery processes. These comments draw upon our extensive experience in addressing AI-related challenges within the scholarly communication ecosystem, including through its use in the research process and by providing high-quality inputs to train, build, and ground new AI systems. STM and its members welcome further collaboration to share our expertise in these areas.

In fact, STM believes that the AI Action Plan should prioritize partnerships between federal agencies and the private sector, including our member publishers. STM believes that the highest priority policy actions should include the following interconnected pillars (1) accelerating research through the use of authoritative vetted and licensed content; (2) ensuring the integrity and provenance of AI outputs; and (3) ensuring the benefits of AI to all by promoting literacy and engagement with AI for all Americans.

Below, we expand on these three priorities with key points and concrete suggestions for action.

1. Accelerating Breakthrough Research

AI is already being used in scholarly publishing to improve the efficiency of research and discovery, to enhance research integrity, and to identify promising areas of new research. It is being deployed internally by publishers to improve the management of articles and peer review, detect potential research misconduct, summarize articles, and improve research recommendations. It is being used externally to support researcher efficiency and help identify promising avenues of study. It seems that the future potential applications in this area are limitless. For example, applications of AI may go beyond testing hypotheses against vast amounts of data to also creating new ones, developing new theories, and exploring new connections.

This innovation builds on and depends upon the availability of high-quality human-created content, vetted by humans, describing intellectual contributions of humans based on experimentation and discovery. The utility of this content for AI and the public is further enhanced by publishers, who validate, normalize, tag and enrich content to make it interoperable and broadly understandable by machines and humans alike. Regardless of the purpose of an AI tool, the accuracy of the scientific record maintained by science and academic publishers is essential to ensure that machine learning has both depth and accuracy.

Therefore, the ability of AI to accelerate breakthrough research depends on intellectual property protection for the functioning of the scholarly communications ecosystem and the continued viability of a scholarly communications industry and ecosystem. The AI Action Plan should ensure:

- **Respect and reinforce existing intellectual property protections**, including but not limited to copyright law, to ensure the ongoing development and availability of high-quality, vetted information and content. We also refer you to an STM submission to the US Patent and Trademark Office in 2020 on the need for intellectual property protection in the context of AI innovation.

- **Leverage and build upon an already existing and well-functioning licensing market**. The market for licensing content to AI developers has developed because the framework for it already exists, both in terms of copyright law and contract law which allows willing parties to agree mutually acceptable terms for direct licensing or via intermediaries. Licensing also fosters accountability, transparency, and provenance with regard to key information such as training materials used by AI systems.

- **Leverage and build upon existing efforts**, including those of STM and our members, to promote high-quality, accurate information and prevent the introduction and spread of "hallucinations" and misinformation.

Specific actions could include:

- The **NAIRR pilot should ensure the use of licensed content** in its activities. Currently, to our knowledge, although NAIRR articulates a need for respect of intellectual property, it does not have explicit requirements related to licensing of content. However, ensuring that AI systems are trained on properly licensed, high-quality data is crucial. By licensing publishers' "Version of Record" (VoR) content for AI training, NAIRR could improve outputs and provide a best practice example of licensing practices.

- Promote **standards that encourage or certify** when AI tools are using **the best quality information**. For scholarly information, this would be the Version of Record which contains the latest peer-reviewed information and is preserved with integrity to include any corrections, to **promote accuracy and reduce the risk of erroneous outputs**.

- **Require audits** and/or certification of the outputs of AI for accuracy and lack of bias, in coordination with the private sector and emerging tools. This could include implementation of corrective mechanisms for inaccurate or outdated information. In the scholarly space, this would ensure that corrections and/or retractions are included in AI, so as **not to waste investments or taxpayer dollars on outdated research** outputs.

2. Build trust with transparency regarding the integrity and provenance of AI outputs

In May 2021, STM published a White Paper outlining principles for an ethical, trustworthy, and human-centric AI. The potential for AI to produce inaccurate information is exacerbated where bad, inaccurate, or false information is used as inputs or in the training sets, rather than the best, high-quality information available. STM and its members are committed to improving the integrity of information and preventing the introduction of information that is inaccurate, unethical, or is unable to be reproduced, including for the use of AI.

Trust in AI systems has been significantly negatively impacted by a lack of clarity about their operation, use of trusted works, and a lack of transparency regarding the works used in training. In addition to improving the reliability and trustworthiness of such systems, transparency obligations are essential to understand the scale and scope of copyrighted works used in AI training, and to enable rightsholders to effectively manage their rights and engage in licensing. This practice will significantly contribute to the development of AI in a legal, ethical and sustainable manner.

Fundamentally, we believe that the following are priorities for an AI Action Plan to make sure that AI and AI outputs are trustworthy, reliable, and trusted:

- **Require public transparency and provenance** of training data and content to enable users to understand and trace back results to their sources and for users and rightsholders to understand what sources were used in training the system. In the scholarly sector, this could include requirements that an AI system and its outputs be placed into a chain of evidence such that results can be more easily reproduced.

- Clarify that **accountability lies in all providers and contributors of AI solutions**, including generative AI models, and throughout the whole life cycle of the AI. Providers should keep all records, documentation, and associated metadata for the duration of the existence of the AI system to support AI accountability throughout the value chain.

- Work to support and fund efforts to **protect the integrity of research and education**, which may be particularly vulnerable to misinformation or misinterpretation of AI outputs. In particular, a government policy should support the development and implementation of defensive mechanisms to identify products (whether quantitative, visual, or textual) created by AI. In the scholarly sector, these would include fake (synthetic) or manipulated image and data detection, identification of paper mills, and the prevention of the use of inaccurate versions of content or illegally sourced content.

Specific actions could include:

- Requiring **a thorough, publicly available list of references and inputs** that enables users or rightsholders to identify what materials were used to train the AI system, or potentially something more sophisticated. The purpose would be to enable users or auditors to track back assertions generated by AIs to the original literature inputs.
- Working with other stakeholders to **identify and safeguard the trustworthiness of AI outputs**. For example, the STM Integrity Hub works to address challenges to the integrity of scholarly communication from human actors as well as AI tools. Through a combination of shared data and experiences, and by harnessing technological innovation, the STM Integrity Hub provides an environment for publishers to collaborate with other parties to develop and operate screening tools, including those using sophisticated AI for the benefit of the entire scholarly ecosystem

3. Enable the benefits of AI to all Americans through education and protection of American creators

Ensuring the development of AI includes all Americans is a key priority for the Trump Administration, as the benefits and opportunities are not necessarily equally distributed across the United States. To make sure that AI is used appropriately, is available for appropriate use by all Americans, and does not undermine American contributions to creativity and innovation, the AI Action Plan should include:

- Promotion of **digital literacy** so that individuals understand the appropriate use and limitations of AI tools.
- Public education on the appropriate use of and risks of use of this information, including the **potential for misinformation** from tools that appear to be reliable but are, in fact, unreliable.
- An **appropriate licensing regime** to protect creators. Such an approach would allow authors or other rightsholders to determine under what conditions and for what purposes their works can be used for training or other purposes. With the current massive uptake of generative AI tools, creators and other rightsholders are seeing their works used for purposes other than what they intended, and even seeing the output of generative AI mimicking or competing with their copyrighted works.

- **Clear terms and conditions** for the use of AI, especially generative AI, so that users do not inadvertently undermine IP.

Specific activities in the Action Plan might include:

- A **public education campaign** that emphasizes AI literacy, technical training, and intellectual property protections.

With appropriate oversight, standards, and regulation, AI is poised to be a benefit to society, if it can be trusted and trustworthy. Trust is at the center of what STM and its members do, as our tagline "advancing trusted research" attests, and therefore STM's members are perfectly positioned to help enable trusted and trustworthy AI. The work that our members do to support digital innovation, from the community's early embrace and stewardship of digital forms of content, tagging and enriching it, creating ontologies, and even using AI to improve it, can be key enablers of a quality AI future. With our long experience with digital technologies, STM and its members stand ready to engage with the federal government in support of trust and integrity in these technologies.

Respectfully submitted,

Dr. Caroline Sutton
CEO
STM

About STM

At STM we support our members in their mission to advance trusted research worldwide. Our more than 140 members collectively publish 66% of all journal articles and tens of thousands of monographs and reference works. As academic and professional publishers, learned societies, university presses, start-ups and established players, we work together to serve society by developing standards and technology to ensure research is of high quality, trustworthy and easy to access. We promote the contribution that publishers make to innovation, openness and the sharing of knowledge and embrace change to support the growth and sustainability of the research ecosystem. As a common good, we provide data and analysis for all involved in the global activity of research.

The majority of our members are small businesses and not-for-profit organizations, who represent tens of thousands of publishing employees, editors, reviewers, researchers, authors, readers, and other professionals across the United States and world who regularly contribute to the advancement of science, learning, culture and innovation throughout the nation. They comprise the bulk of a $25 billion publishing industry that contributes significantly to the U.S. economy and enhances the U.S. balance of trade.